SUPPORTING STUDY

Ambulance Reachability and the Golden Hour

Road accidents on Indian National highways, ambulance reachability and transportation of injured to trauma facility: Survey-based introspection of golden hour

Journal of Family Medicine and Primary Care | PubMed ID: 38605810 | https://pubmed.ncbi.nlm.nih.gov/38605810/

Background and Context

India has pushed hard to build and upgrade its road infrastructure in recent years. Better highways, however, do not automatically mean safer ones. The article's starting point is that highways often lack enough ambulance services and trauma facilities along their length. When help is far away, accidents turn into a steady loss of lives and serious injuries, and everyday road users are exposed to greater risk.

The study is framed around the **golden hour**, the idea that the first hour after an injury largely decides a critically injured person's chance of survival. Indian road-safety literature often points out how large the problem is: national figures cited in other studies report more than 4 lakh road accidents and over 1.4 lakh deaths in 2022, which works out to dozens of accidents and many deaths every hour. In this setting, every minute an ambulance spends reaching the spot or reaching a hospital matters.

Aim and Method

The authors set out to measure two things on national highways that carry heavy commercial traffic:

- how long an ambulance takes to **reach the accident spot** (response time), and
- how long it then takes to **transport the injured person to the nearest trauma facility**.

They used a survey-based design. A questionnaire was built with variables that record and assess these two time intervals, so that the whole journey from the accident to trauma care could be looked at, not just the ambulance's arrival.

Key Findings

The survey led the authors to sort highway accidents into two categories:

- **Category 1:** accidents on highways near city limits or dense settlements.
- **Category 2:** accidents on core highways, away from settlements.

Accidents on highways adjacent to or passing through city limits and dense settlements were reported to be more frequent than those on core highways. This matters because these stretches mix fast long-distance traffic, heavy commercial vehicles, local vehicles and pedestrians, and are also the places where congestion can slow down an ambulance.

Conclusion of the Authors

The authors argue that technology-backed reforms are strongly needed to turn Indian highways into **'emergency rescuable highways'**. Their message is that a safe and sustainable transport system must be able to get help to victims quickly, not just prevent crashes.

A reader's letter published in response to the article adds useful points: highway quality, traffic jams, ambulance availability and how quickly calls are answered are all important when judging how fast an ambulance can reach an accident spot.

Why We Used This Article in Our Case Study

Our project tackles two linked problems: congestion caused by fixed-timing signals, and slow emergency response. This article gave us evidence for the second one.

- It shows that delayed ambulance arrival is a documented problem in India, not just an assumption, which supports the need behind our Emergency Vehicle Priority System.
- It finds that accidents are more frequent near city limits and dense settlements, the same busy urban areas our signal system is designed for.
- It calls for technology-backed reform, which is exactly the approach of our AI, sensor and IoT solution.
- It is a peer-reviewed, India-specific source, which strengthens the credibility of our rationale.

What We Learned from It

- **Time is survival.** The golden hour means every minute lost in traffic can affect whether an injured person lives.
- **Response has two stages.** Reaching the spot and then reaching a trauma facility can both be delayed, and congestion affects both.
- **Location matters.** City-edge and dense-settlement roads are high-risk zones, so they are sensible places to pilot a smart traffic system first.
- **Infrastructure alone is not enough.** New roads need emergency response built into them, and technology can help provide that.
- **Many factors affect response.** Road quality, jams, ambulance availability and call handling all play a part, so our system is one piece of a larger solution.
- **Link to our project.** Smart signals reduce congestion and pollution for everyone, while automatic green-light priority for ambulances helps give victims their best chance within the golden hour.